---

1. **Obtén la dirección IP de los siguientes dominios: www.uhu.es, www.us.es, es.wikipedia.org.**

Para uhu:

```
fran@FranDesplie:~$ dig www.uhu.es

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.uhu.es
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50187
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.uhu.es.                    IN      A

;; ANSWER SECTION:
www.uhu.es.             281     IN      A       150.214.167.13

;; Query time: 16 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:36:42 UTC 2026
;; MSG SIZE  rcvd: 55
```

Para us.es:

```
fran@FranDesplie:~$ dig www.us.es

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.us.es
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 1570
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.us.es.                     IN      A

;; ANSWER SECTION:
www.us.es.              14159   IN      A       193.147.175.38

;; Query time: 18 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:37:37 UTC 2026
;; MSG SIZE  rcvd: 54
```

Para Wikipedia:

```
fran@FranDesplie:~$ dig es.wikipedia.org

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> es.wikipedia.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60925
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;es.wikipedia.org.              IN      A

;; ANSWER SECTION:
es.wikipedia.org.       7258    IN      CNAME   dyna.wikimedia.org.
dyna.wikimedia.org.     180     IN      A       185.15.58.224

;; Query time: 29 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:38:01 UTC 2026
;; MSG SIZE  rcvd: 90
```

2. Obtén la dirección y los servidor DNS que corresponden a los siguientes dominios: net, com, us.es, wikipedia.org.

net NS:

```
fran@FranDesplie:~$ dig net NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> net NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 38367
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;net.                            IN      NS

;; ANSWER SECTION:
net.                    21600   IN      NS      k.gtld-servers.net.
net.                    21600   IN      NS      d.gtld-servers.net.
net.                    21600   IN      NS      e.gtld-servers.net.
net.                    21600   IN      NS      l.gtld-servers.net.
net.                    21600   IN      NS      a.gtld-servers.net.
net.                    21600   IN      NS      b.gtld-servers.net.
net.                    21600   IN      NS      j.gtld-servers.net.
net.                    21600   IN      NS      i.gtld-servers.net.
net.                    21600   IN      NS      h.gtld-servers.net.
net.                    21600   IN      NS      f.gtld-servers.net.
net.                    21600   IN      NS      c.gtld-servers.net.
net.                    21600   IN      NS      m.gtld-servers.net.
net.                    21600   IN      NS      g.gtld-servers.net.

;; Query time: 41 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:40:01 UTC 2026
;; MSG SIZE  rcvd: 253
```

com NS:

```
fran@FranDesplie:~$ dig com NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33211
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;com.                            IN      NS

;; ANSWER SECTION:
com.                    21600   IN      NS      d.gtld-servers.net.
com.                    21600   IN      NS      h.gtld-servers.net.
com.                    21600   IN      NS      c.gtld-servers.net.
com.                    21600   IN      NS      l.gtld-servers.net.
com.                    21600   IN      NS      m.gtld-servers.net.
com.                    21600   IN      NS      k.gtld-servers.net.
com.                    21600   IN      NS      e.gtld-servers.net.
com.                    21600   IN      NS      g.gtld-servers.net.
com.                    21600   IN      NS      f.gtld-servers.net.
com.                    21600   IN      NS      j.gtld-servers.net.
com.                    21600   IN      NS      i.gtld-servers.net.
com.                    21600   IN      NS      a.gtld-servers.net.
com.                    21600   IN      NS      b.gtld-servers.net.

;; Query time: 42 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:40:38 UTC 2026
;; MSG SIZE  rcvd: 256
```

us.es NS:

```
fran@FranDesplie:~$ dig us.es NS

 <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> us.es NS
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25838
; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
 EDNS: version: 0, flags:; udp: 65494
; QUESTION SECTION:
us.es.                          IN      NS

; ANSWER SECTION:
s.es.                   14156   IN      NS      sun.rediris.es.
s.es.                   14156   IN      NS      onix.us.es.
s.es.                   14156   IN      NS      dns1.cica.es.
s.es.                   14156   IN      NS      jade.us.es.
s.es.                   14156   IN      NS      chico.rediris.es.
s.es.                   14156   IN      NS      dns2.cica.es.

; Query time: 17 msec
; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
; WHEN: Sat Feb 07 19:40:55 UTC 2026
; MSG SIZE  rcvd: 161
```

wikipedia.org NS:

```
fran@FranDesplie:~$ dig wikipedia.org NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> wikipedia.org NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29587
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;wikipedia.org.                 IN      NS

;; ANSWER SECTION:
wikipedia.org.          1848    IN      NS      ns1.wikimedia.org.
wikipedia.org.          1848    IN      NS      ns2.wikimedia.org.
wikipedia.org.          1848    IN      NS      ns0.wikimedia.org.

;; Query time: 13 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:41:17 UTC 2026
;; MSG SIZE  rcvd: 106
```

3. Averigua los registros MX de los siguientes dominios: uhu.es, us.es, wikipedia.org.

uhu.es MX:

```
fran@FranDesplie:~$ dig uhu.es MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> uhu.es MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 65038
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;uhu.es.                                IN      MX

;; ANSWER SECTION:
uhu.es.                 3600    IN      MX      10 ms56712221.msv1.invalid.
uhu.es.                 3600    IN      MX      10 mx04.puc.rediris.es.
uhu.es.                 3600    IN      MX      10 mx03.puc.rediris.es.
uhu.es.                 3600    IN      MX      10 uhu.es.

;; Query time: 50 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:43:01 UTC 2026
;; MSG SIZE  rcvd: 144
```

us.es MX:

```
fran@FranDesplie:~$ dig us.es MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> us.es MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55305
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;us.es.                         IN      MX

;; ANSWER SECTION:
us.es.                  346     IN      MX      10 buzon.us.es.

;; Query time: 16 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:43:19 UTC 2026
;; MSG SIZE  rcvd: 56
```

wikipedia.org MX:

```
fran@FranDesplie:~$ dig wikipedia.org MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> wikipedia.org MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59490
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;wikipedia.org.                 IN      MX

;; ANSWER SECTION:
wikipedia.org.          300     IN      MX      10 mx-in1001.wikimedia.org.
wikipedia.org.          300     IN      MX      10 mx-in2001.wikimedia.org.

;; Query time: 32 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:43:38 UTC 2026
;; MSG SIZE  rcvd: 104
```

4. Obtén la dirección IPV6 de www.isc.org.

```
fran@FranDesplie:~$ dig www.isc.org AAAA

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.isc.org AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3394
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.isc.org.                   IN      AAAA

;; ANSWER SECTION:
www.isc.org.            300     IN      CNAME   isc.map.fastlydns.net.
isc.map.fastlydns.net.  60      IN      AAAA    2a04:4e42::729
isc.map.fastlydns.net.  60      IN      AAAA    2a04:4e42:600::729
isc.map.fastlydns.net.  60      IN      AAAA    2a04:4e42:400::729
isc.map.fastlydns.net.  60      IN      AAAA    2a04:4e42:200::729

;; Query time: 222 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:45:06 UTC 2026
;; MSG SIZE  rcvd: 187
```

5. Muestra los servidores de correo de yahoo.com.

```
fran@FranDesplie:~$ dig yahoo.com MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> yahoo.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 36521
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;yahoo.com.                     IN      MX

;; ANSWER SECTION:
yahoo.com.              1086    IN      MX      1 mta6.am0.yahoodns.net.
yahoo.com.              1086    IN      MX      1 mta5.am0.yahoodns.net.
yahoo.com.              1086    IN      MX      1 mta7.am0.yahoodns.net.

;; Query time: 17 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:46:36 UTC 2026
;; MSG SIZE  rcvd: 117
```

6. Muestra la información asociada con la dirección 75.126.153.206.

```
fran@FranDesplie:~$ dig -x 75.126.153.206

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> -x 75.126.153.206
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24953
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;206.153.126.75.in-addr.arpa.   IN      PTR

;; ANSWER SECTION:
206.153.126.75.in-addr.arpa. 3600 IN    PTR     ce.99.7e4b.ip4.static.sl-reverse.com.

;; Query time: 45 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:48:18 UTC 2026
;; MSG SIZE  rcvd: 106
```

7. Muestra la dirección de www.google.es utilizando uno de los servidores DNS de google.com.

```
fran@FranDesplie:~$ dig @8.8.8.8 www.google.es

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @8.8.8.8 www.google.es
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8930
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.google.es.                 IN      A

;; ANSWER SECTION:
www.google.es.          87      IN      A       172.217.20.227

;; Query time: 13 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Feb 07 19:49:51 UTC 2026
;; MSG SIZE  rcvd: 58
```

8. Muestra información sobre el TTL de drive.google.com. Ejecútalo en varias ocasiones y comprueba cómo cambia el valor.

```
fran@FranDesplie:~$ dig drive.google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> drive.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22113
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;drive.google.com.              IN      A

;; ANSWER SECTION:
drive.google.com.       300     IN      A       142.251.142.142

;; Query time: 35 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:50:44 UTC 2026
;; MSG SIZE  rcvd: 61
```

9. Muestra los registro type NS de redhat.com.

```
fran@FranDesplie:~$ dig redhat.com NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> redhat.com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21136
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;redhat.com.                    IN      NS

;; ANSWER SECTION:
redhat.com.             3600    IN      NS      dns2.p02.nsone.net.
redhat.com.             3600    IN      NS      dns4.p01.nsone.net.
redhat.com.             3600    IN      NS      dns3.p01.nsone.net.
redhat.com.             3600    IN      NS      dns1.p01.nsone.net.
redhat.com.             3600    IN      NS      dns2.p01.nsone.net.
redhat.com.             3600    IN      NS      dns1.p02.nsone.net.

;; Query time: 30 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:51:25 UTC 2026
;; MSG SIZE  rcvd: 170
```

10. Muestra todos los registros de redhat.

```
fran@FranDesplie:~$ dig redhat.com ANY

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> redhat.com ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31313
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;redhat.com.                    IN      ANY

;; ANSWER SECTION:
redhat.com.             3600    IN      HINFO   "RFC8482" ""

;; Query time: 30 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (TCP)
;; WHEN: Sat Feb 07 19:52:41 UTC 2026
;; MSG SIZE  rcvd: 60
```

11. Crea un fichero de texto que contenga los dominios.

```
fran@FranDesplie:~$ echo -e "www.uhu.es\nwww.us.es\mes.wikipedia.org\nredhat.com" > archivo_dominios.txt
fran@FranDesplie:~$ cat archivo_dominios.txt
www.uhu.es
www.us.es\mes.wikipedia.org
redhat.com
fran@FranDesplie:~$ _
```

12. Haz una consulta con dig empleando el fichero anterior.

```
fran@FranDesplie:~$ dig -f archivo_dominios.txt | less_
```

Como el resultado era mucha información, tuve que hacer less para poder luego moverme con las flechas del teclado arriba y abajo para ver toda la información, aquí esta toda la que devolvió:

```
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.uhu.es
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 9026
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.uhu.es.                    IN      A

;; ANSWER SECTION:
www.uhu.es.             859     IN      A       150.214.167.13

;; Query time: 2 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:57:20 UTC 2026
;; MSG SIZE  rcvd: 55

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.us.es\mes.wikipedia.org
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 10923
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.us.esmes.wikipedia.org.    IN      A

;; AUTHORITY SECTION:
wikipedia.org.          1738    IN      SOA     ns0.wikimedia.org. hostmaster.wikimedia.org. 2026012613 43200 7200 1209

;; Query time: 20 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:57:20 UTC 2026
;; MSG SIZE  rcvd: 116
```

```
; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> redhat.com
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2385
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;redhat.com.                    IN      A

;; ANSWER SECTION:
redhat.com.             690     IN      A       34.235.198.240
redhat.com.             690     IN      A       52.200.142.250

;; Query time: 13 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Feb 07 19:57:20 UTC 2026
;; MSG SIZE  rcvd: 71

(END)
```